GGG-H-211D
July 19, 1985
SUPERSEDING
GGG-H-211C
May 19, 1976

FEDERAL SPECIFICATION

HELMETS, WELDERS'; HANDSHIELD, WELDING; AND PLATES, WELDING

This specification is approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers welders helmets, handshields, and plates.

1.2 Classification. *The* helmets, handshield, and plates shall be of the following types, styles, and classes, as specified (see 6.2):

Type I - Helmets

Style a - Curved or bib front
class 1 - Fixed front
class 2 - Lift front

Style b - Straight front
class 1 - Fixed front
Class 2 - Lift front

Style c - Narrow front
Class 1 - Fixed front
Class 2 - Lift front

Type II - Handshield

Type III - Plates

Style a - Filter
Style b - Cover
Style c - Safety

FSC 4240

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-566	Boxes, Folding, Paperboard
PPP-B-576	Box, Wood, Cleated, Veneer, Paper Overlaid
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-591	Boxes, Shipping, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-B-665	Boxes: Paperboard, Metal Edged and Components
PPP-B-676	Boxes, Setup
PPP-T-76	Tape, Packaging, Paper (For Carton Sealing)

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies)

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

(Single copies of this specification, other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Washington, DC; Philadelphia, PA; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications:

MIL-P-116 - Preservation, Methods of
MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of Military specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply-

American National Standards Institute, Inc., (ANSI) Standard:

Z87.I - Occupational and Educational me and Face Protection

(Application for copies should be addressed to American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

ASTM Standards:

D 635 - Rate of Burning and/or Extent and Time of Burning of Self-Supporting Plastics in a Horizontal Position
D 1003 - Haze and Luminous Transmittance of Transparent Plastics
D 3951 - Commercial Packaging

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Compliance with ANSI Z87.1. Helmets, handshields, and plates shall comply with the requirements of ANSI 287.1 except as otherwise specified herein. The contractor shall certify that this requirement has been met.

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3.2 Shell material.

3.2.1 Characteristics. All shell material in helmets and handshields shall be thermally insulating, noncombustible or slow burning, and opaqued to ultraviolet, infrared, and visible light radiations. The contractor shall certify that these requirements have been met.

3.2.2 Dimensions. The shell material shall be made of vulcanized fiber not less than 1.2 millimeter (mm) thick, or reinforced plastic or other material not less than 1.0 mm thick.

3.2.3 Burn rate. The shell material shall not burn at a rate greater than 75 mm per minute when tested as specified in 4.2.4.1.

3.3 Contour and dimensions. The body of the helmets and handshield shall conform to the contour and dimensions shown on figures 1, 2, 3 and 4. Provided these requirements are met, the actual appearance of the body of the helmets and handshield may vary from that shown in the figures.

3.4 Weight. Helmets and handshields, exclusive of plates, shall not weigh more than the following:

Type I, class 1, or type II	680 grams (g)
Type I, class 2	793 g

3.5 Light leakage. The helmets and handshields shall be so constructed that they will be free from light leakage when tested as specified in 4.2.4.2.

3.6 Electrical insulation. Any metal parts, including rivets, which extend through the shell of the helmet or handshield and which might touch the wearer's head or face, shall not be exposed on the inside of the helmet or handshield. Such metal parts shall be insulated by a covering of nonconductive insulating material, other than lacquer or varnish, which completely insulates and covers such metal parts. The insulation shall be sufficient to withstand the insulation test specified in 4.2.4.3.

3.7 Finish. All interior and exterior parts shall have a permanent, nonreflecting finish and shall be smooth and free from sharp edges or any irregularities which might present a hazard to the wearer or user.

3.8 Identification marking. Helmets, handshields, and style a and style c plates shall bear a permanent distinctive marking identifying the manufacturer. style a plates shall be marked with the shade number (see 3.11.2.2). Heat-treated style c plates shall be marked with the letter "H."

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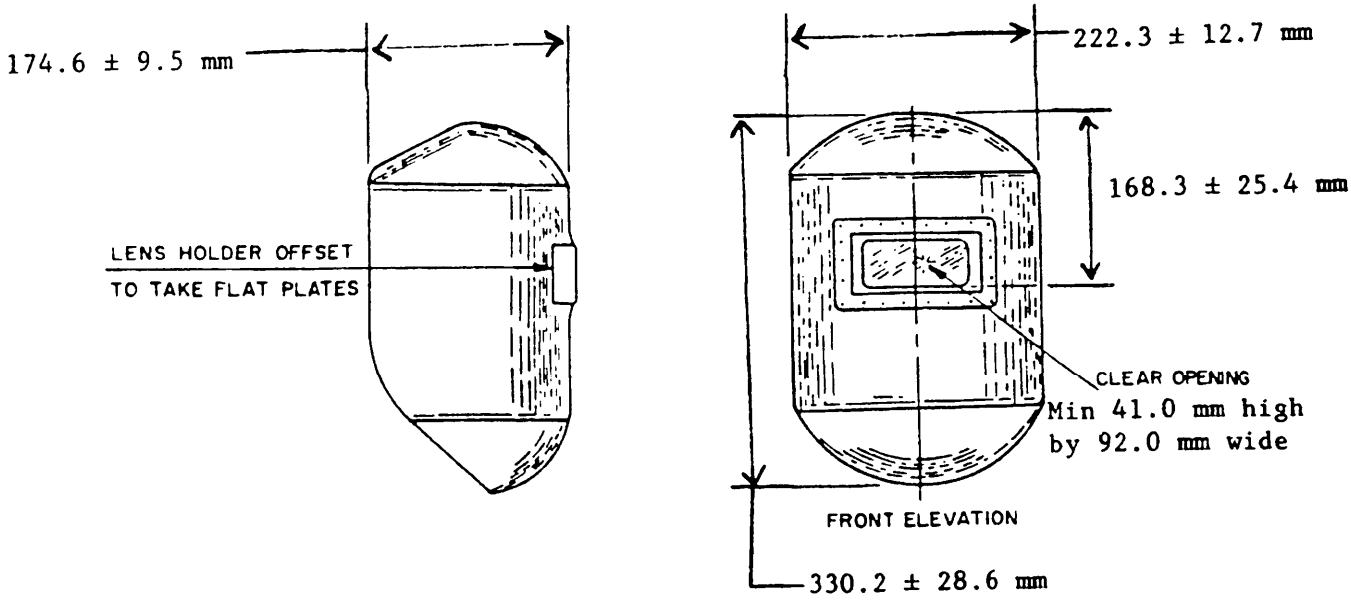


FIGURE 1. Type I, style a, curved or bib front helmets

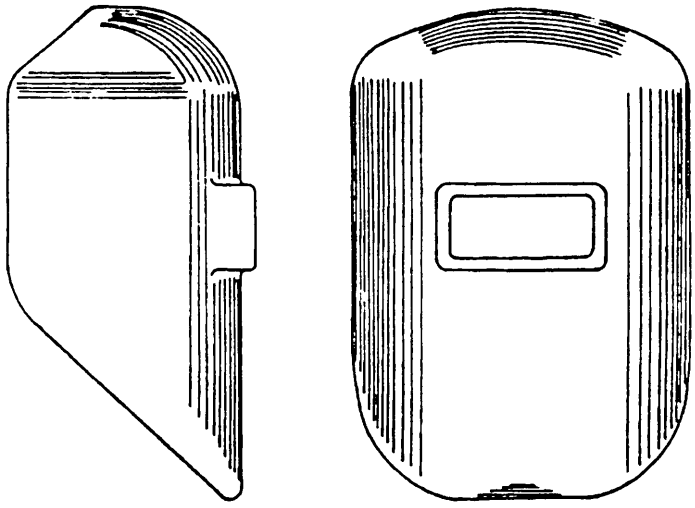


FIGURE 2. Type I, style b, straight front helmets.
For dimensions see figure 1.

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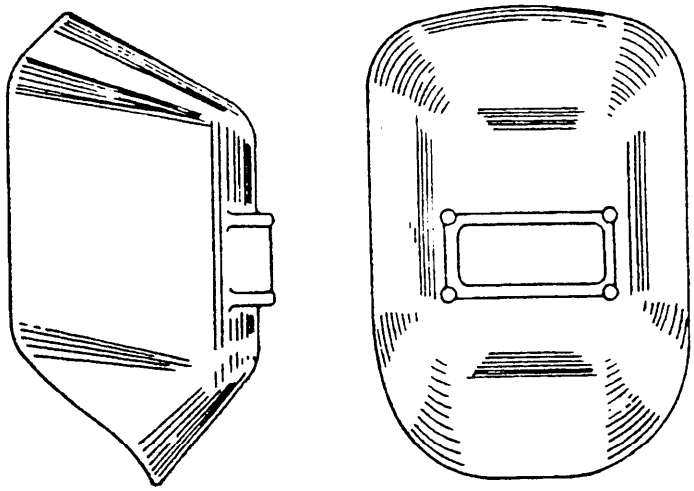


FIGURE 3. Type I, style c, narrow front helmets . For dimensions see figure 1.

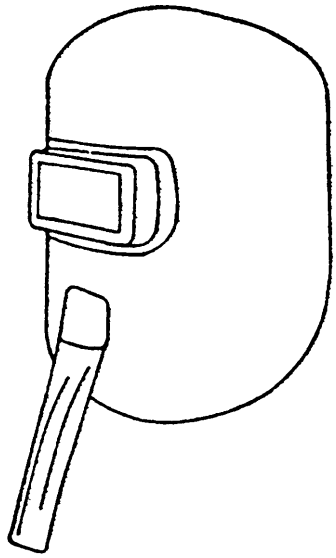


FIGURE 4. Handshields, type II. For dimensions see figure 1.

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3.9 Type I helmets.

3.9.1 All classes.

3.9.1.1 Warpage resistance. Helmets shall be constructed so as to be resistant to warpage without a change in the original dimensions of more than 2.0 percent (see figures 1, 2, and 3) when tested as specified in 4.2.4.4.

3.9.1.2 Headgear. The headgear shall retain the helmet. comfortably and firmly in place *on* the wearer's head, and shall be so constructed that-the helmet may be tilted back over the head. The headgear shall be adjustable so as to permit positioning of the style a and style c plates in the line of vision of the wearer. The contractor shall certify that these requirements have been met.

3.9.1.3 Sweatband. The headgear shall be fitted with a removable and replaceable sweatband covering at least the forehead portion of the headband. The sweatband shall be made of leather or other material which is slow burning, non-irritating, and non-allergenic to normal skin. The contractor shall certify that these requirements have been met.

3.9.1.4 Headband. The headband shall be readily adjustable to various head sizes without the use of tools, and shall hold firmly without slippage after being adjusted. Adjustment shall be by means of a positive locking arrangement *or* ratchet adjustment type. Adjustment mechanism and movements, together with that portion of the band *over* which the ends travel, shall be continuous (not stepwise) and shall be enclosed within a sleeve or shall be otherwise protected to prevent snagging the hair while wearing or making adjustments. The contractor shall certify that these requirements have been met.

3.9.1.5 Chin rest or position stop. Helmets shall be provided with either a chin rest or position stop. Chin rest shall extend inside the helmet, across the bottom, and shall be thermally nonconductive. The strip shall be arched to provide additional strength. The chin rest shall be connected to the helmet in such a manner that it may be adjusted to position the helmet in front of the face. An adjustable position stop in lieu of chin rest is acceptable, provided it performs the same function as the chin rest. The contractor shall certify that these requirements have been met.

3.9.2 Class 1, fixed front. The helmet shall be fitted with a style a plate mounting with a clear opening no less than 41 mm high and 92 mm wide that will hold the style a and style b plates securely and firmly in place. Plate mountings shall be such as to permit, without the use of tools, the removal and replacement of plates without damage to the plates or mounting. The mounting frame may be of metal or thermosetting or therm-plastic material. The mounting frame shall be securely mounted to, or form an integral part of, the helmet or handshield. Insulation requiring removal in order to replace plates shall not be used. The contractor shall certify that these requirements have been met.

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3.9.3 class 2, lift front. The lift front assembly shall consist of two parts: the stationary portion, which may be fabricated or integrally molded, and a lift portion which may be fabricated from thermosetting or thermoplastic material, die cast aluminum, magnesium alloy, stamped drawn-metal or other material specifically approved by the contracting activity. The lift portion shall be so constructed that it will stay up or down, but will not remain in a partially opened position (see figure 5). The lift front shall be designed to accommodate three plates: A style c plate in the stationary portion, and a style b and style a plate in the lift part. Unless otherwise specified (see 6.2), lift front helmets shall be furnished without plates. *The* lift front shall permit, without the use of tools, the removal and replacement of each of the three plates specified herein. The contractor shall certify that these requirements have been met.

309.3.1 Light leakage. The lift front seal against the helmet shall be free from light leakage when tested as specified in 4.2.4.2.

3.10 Type II, handshields.

3.10.1 Handle. The handle *cross* section shall be securely fitted at the lower part of the handshield. It shall be made of thermally nonconductive material at least 32 mm in diameter with a wall not less than 1.27 mm thick. The handle may be round, oval or triangularly shaped, and shall extend at least 130 mm below the point of attachment to the handshield. It shall be finished smooth, knurled, or fluted. The contractor shall certify that these requirements have been met.

3.11 Type III plates.

3.11.1 All style plates.

3.11.1.1 Optical defects. Type III plates shall be free from striae waves, flows, or any defects which may impair the optical quality, when tested as specified in 4.2.4.5.

3.11.1.2 Definition. Type 111 plates shall be tested for definition in accordance with 4.2.4 .6. As a test chart, the resolving power chart pattern 20 specified in ANSI Z87.1 shall be clearly resolved with the target placed at a distance of 10.7 meters from the telescope objective used for testing the plates.

3.11.1.3 Prismatic power. Prismatic effects of type III plates shall not exceed 1/16 prism diopter approximately 2.2 minutes of angular deviation), when tested as specified in 4.2.4.7.

3.11.1.4 Refractive power and astigmatism. Type III plates shall not have refractive power in excess of ± 1/16 diopter and the astigmatism (difference in refractive *power* of any two meridians) shall not exceed 1/16 diopter when tested as specified in 4.2.4.8.

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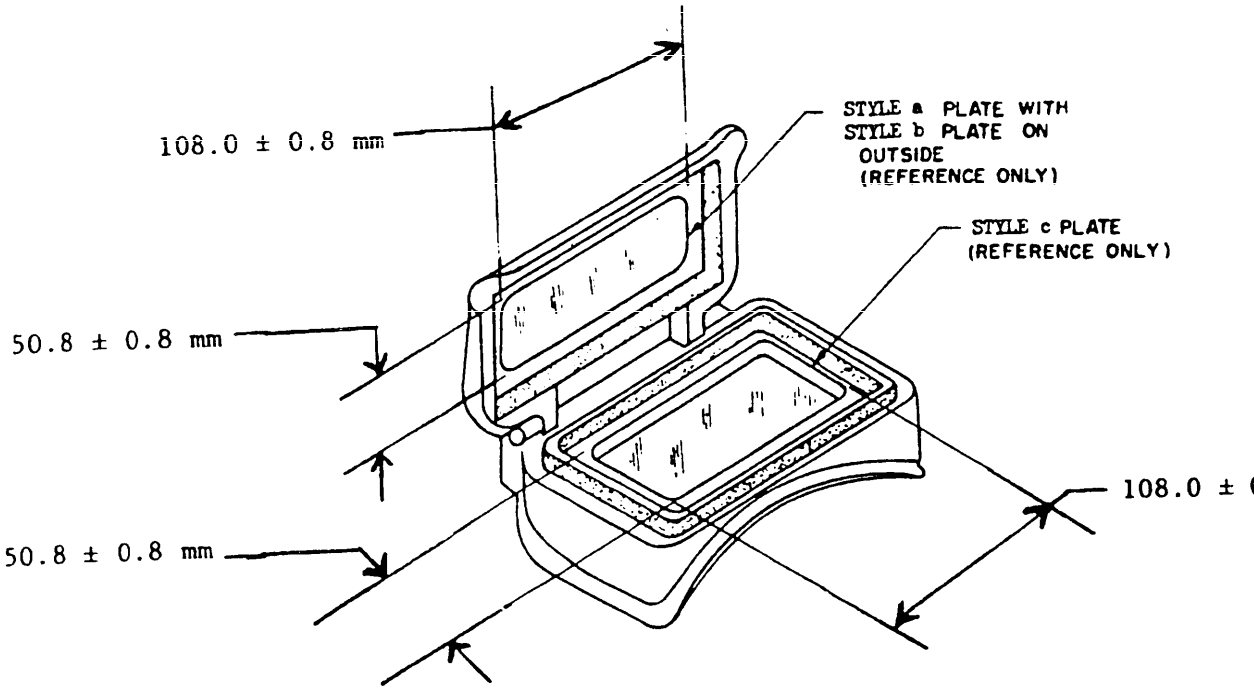


FIGURE 5. Typical type I, class 2, lift front

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3.11.1.5 Flame propagation rate (plastic plates only). All plastic materials used in the construction of style a, b or c plates shall be noncombustible or slow burning *when* tested as specified in 4.2.4.9.

3.11.1.6 Thermal shock. Optical surfaces of type III plates shall not delaminate or deteriorate or there shall be no fracture of glass when the plates are tested for thermal shock as specified in 4.2.4.10.

3.11.1.7 Workmanship. The optical surfaces of the plates shall be free from surface defects to within 9.5 mm of the edge. The optical surfaces shall be flat and substantially parallel. The corners of the edges shall be ground sufficiently to prevent injury to the user.

3.11.2 Style a plates.

3.11.2.1 Physical characteristics. Style a plates shall be made of single sheet glass, plastic, laminated glass *or* other material, and shall measure 50.8 ± 0.8 mm wide, 108.0 ± 0.8 mm long, and between 2.0 mm and 3.8 mm thick.

3.11.2.2 Radiant energy transmittance. Style a plates shall be of the shade specified (see 6.2) and shall conform to the radiant energy transmittance requirements shown in table I, when tested as specified in 4.2.4.11.

3*11.2.3 Impact resistance. Style a plates shall not fracture when tested for impact resistance as specified in 4.2.4.12.

3.11.2.4 Maximum transparency. The maximum transparency of style a plates shall be between 490 and 600 nanometers gradually decreasing on either side. Plates conforming to this requirement shall have a dominant hue ranging from bluish-green, yellowish-green to greenish-yellow, depending upon the composition of the filter, when tested as specified in 4.2.4.13.

3.11.2.5 Haze. Plastic of all types used in style a plates shall exhibit not more than 6.0 percent haze when tested as specified in 4.2.4.14.

3.11.2.6 Distance from plate to eye. The distance from style a plates to eye when the plates are mounted in helmets shall be not less than 50.8 mm. The contractor shall certify that this requirement has been met.

3.11.3 Style b plates.

3.11.3.1 Physical characteristics. Style b plates shall be single, solid glass plates either uncoated, plastic coated on one or both sides, or shall be of allyl-plastic sheet, as specified (see 6.2). Allyl plastic sheets shall be allyl cast plastic CR-39 (Columbia Resin 39) or equal. All style b plates shall be the same peripheral size and shape as style a plates. The contractor shall certify that these requirements have been met. Additionally, the thickness of the coated or uncoated style b plates shall be not less than 1.2 mm nor more than 3.8 mm.

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TABLE I. Transmittance and tolerances in transmittance of various shades of plates

Shade No.	Density for visible radiation			Percent transmittance of total visible*			Percent maximum total infrared	Percent maximum ultraviolet spectral transmittance, Nanometers			
	Min.	Standard	Max.	Min.	Standard	Max.		313	334	365	405
1.5	0.17	0.214	0.26	55	61.5	67	25	0.2	0.8	25	65
1.7	0.26	0.300	0.36	43	50.1	55	20	0.2	0.7	20	50
2.0	0.36	0.429	0.54	29	37.3	43	15	0.2	0.5	14	35
2.5	0.54	0.643	0.75	18.0	22.8	29	12	0.2	0.3	5	15
3.0	0.75	0.857	1.07	8.50	13.9	18.0	9.0	0.2	0.2	0.5	6
4.0	1.07	1.286	1.50	3.16	5.18	8.50	5.0	0.2	0.2	0.5	1.0
5.0	1.50	1.714	1.93	1.18	1.93	3.16	2.5	0.2	0.2	0.2	0.5
6.0	1.93	2.143	2.36	0.44	0.72	1.18	1.5	0.1	0.1	0.1	0.5
7.0	2.36	2.571	2.79	0.164	0.27	0.44	1.3	0.1	0.1	0.1	0.5
8.0	2.79	3.000	3.21	0.061	0.100	0.164	1.0	0.1	0.1	0.1	0.5
9.0	3.21	3.429	3.64	0.023	0.037	0.061	0.8	0.1	0.1	0.1	0.5
10.0	3.64	3.854	4.07	0.0085	0.0139	0.023	0.6	0.1	0.1	0.1	0.5
11.0	4.07	4.286	4.50	0.0032	0.0052	0.0085	0.5	0.05	0.05	0.05	0.1
12.0	4.50	4.714	4.93	0.0012	0.0019	0.0032	0.5	0.05	0.05	0.05	0.1
13.0	4.93	5.143	5.36	0.00044	0.00072	0.0012	0.4	0.05	0.05	0.05	0.1
14.0	5.36	5.571	5.79	0.00016	0.00027	0.00044	0.3	0.05	0.05	0.05	0.1

*Visible transmittance is based on the percentage of the visible radiation of a gas-filled tungsten lamp operated at normal rating, determined photometrically.

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3.11.3.2 Visible transmittance. The uncoated style b plates shall transmit not less than 89 percent of incident visible light; the plastic-coated style b plates shall transmit not less than 75 percent of incident visible light, and the plastic style b plates shall transmit not less than 86 percent of incident visible light when tested as specified in 4.2.4.11.

3.11.3.3 Water resistance. The style b plate plastic coating shall be resistant to water when tested as specified in 4.2.4.15.

3.11.4 Style c plates.

3.11.4.1 Physical characteristics. Style c plates shall be the same peripheral size and shape as the style a and style b plates, and shall be clear and either heat-treated glass or plastic. The contractor shall certify that these requirements have been met.

3.11*4.2 Dimensions. Glass plates shall be not less than 3.0 mm thick nor more than 3.8 mm thick. Plastic plates shall be not less than 1.2 mm thick.

3.11.4$_0$3 Impact resistance. Style c plates shall not fracture when tested for impact resistance as specified in 4.2.4.12.

3.11.4.4 Penetration. Plastic style c plates shall not be pierced through when tested for resistance to penetration as specified in 4.2.4.16.

3.12 Workmanship. The helmets, handshields, and plates shall have no sharp edges or burrs and shall be free from any other defects which detract from their appearance or which may impair their proper usage or serviceability

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of all items of one type, class, and style, offered for delivery at one time.

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4.2.2 Sampling.

4.2.2.1 For examination. A random sample of helmets, handshields, or plates, as applicable, shall be selected from each lot in accordance with MIL-STD-105, inspection level II.

4.2.2.2 For test. A random sample of helmets, handshields, or plates, as applicable, shall be selected from each lot for testing in accordance with MIL-STD-105, inspection level S-4.

4.2.3 Inspection procedure.

4.2.3.1 For examination. Each of the sample helmets, handshields, or plates selected in accordance with 4.2.2.1, and the packaging thereof shall be examined in accordance with MIL-STD-105 and the following classification of defects:

(a) Helmets, handshield, and plates.

Categories		Defects
Critical:		None defined
Major	AQL	1.5 percent defective
101		Shell material dimensions not as specified (see 3.2.2)
102		Contour and dimensions of helmets and handshields not as specified in figures 1, 2, 3, or 4, as applicable (see 3.2)
103		Weight of helmet *or* handshield not as specified (see 3.4)
104		Metal parts extending through the shell of the helmet or handshield exposed on inside surface (see 3.6)
105		Electrical insulation material covering metal parts missing or not as specified (see 3.6)
106		Finish missing or not as specified (see 3.7)
107		Identification marking missing or not as specified (see 3.8)
108		Helmet opening dimensions not as specified (see 3.9.2)
109		Handshield handle not as specified (see 3.10.1)
110		Workmanship (see 3.11.1.7 and 3.12)
111		Style a plate dimensions not as specified (see 3.11.2.1)
112		Shade of style a plates not as specified (see 3.11.2.2)
113		Thickness of style b plates not as specified (see 3.11.3.1)
114		Style c plate dimensions not as specified (see 3.11.4.2)

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(b) Packaging (see section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Contents per container not as specified
102	Container not as specified
103	*Container closure* not as specified
104	Container damaged
105	Helmet or handshield not clean (see 5.1.1.1)
106	Gross weight of intermediate boxes for plates exceeds 9.1 kilograms (see 5.1.1.4)
107	Caseliners in shipping containers for level A pack missing or not as specified when required (see 5.2.1)
108	Gross weight of wood boxes for pack exceeds 91 kilograms (see 5.2.1)
109	Marking incorrect, missing, or illegible

4.2.3.2 For test. Sample helmets, handshields, and plates shall be tested as specified in the applicable portions of 4.2.4 using an AQL of 1.0 percent defective for the light leakage test and *an* AQL of 2.5 percent defective for the other tests.

4.2.4 Tests. Tests shall be conducted as follows:

4.2.4.1 Burn rate. Determine the burn rate in accordance with ASTM D 635 using three strips, 13 mm wide and 130 mm long, cut from the flat area of one helmet, shell or handshield as the test specimen in lieu of the 10 specimens required in ASTM D 635. The burn rate of the vulcanized fiber material shall be reported as the average value obtained for the three specimens tested. Material of any other composition shall have all three test specimens rated as slow burning by this test.

4.2.4.2 Light leakage. Hold the helmet *or* handshield up to a projection type lamp No. T-8 or other high-powered, gas-filled, tungsten-filament, electric incandescent lamp and observe for any light leakage.

4.2.4.3 Electrical insulation. Connect one terminal *of* a 1,000-volt (root-mean-square value), alternating current, 60-Hertz source to the insulated metallic parts being tested inside the helmet or handshield. Connect the other terminal to a roving electrode and test each exterior metallic projection for adequate insulation: Any evidence of electrical conductivity or breakdown shall be considered as unsatisfactory and shall be cause for rejection.

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4.2.4.4 Warpage resistance. Place the helmet or handshield previously measured for conformance with the requirements shown on figure 1 in an oven heated to 66° ± 1°C for 24 hours. Allow the sample to stand for 4 hours under normal atmospheric conditions and again measure the dimensions.

4.2.4.5 Optical defects. Test plates for effective striae, prismatic effect, focus, or astigmatism, and quality of optical surface as indicated by clearness of image viewed through the style a or style b plates when placed in front of a telescope, using the applicable procedures described in ANSI Z87.1.

4.2.4.6 Definition. Test the plates for definition with a 8-power telescope which has an effective aperture of 19.05 mm focused at a distance of 10.7 meters on the illuminated resolving power chart pattern 20 specified in ANSI Z87.1.

4.2.4.7 Prismatic power. The plates shall be tested for prismatic power with an 8-power telescope which has an effective aperture of 19.05 mm and is equipped with cross hairs in the focal plane of the ocular. The telescope shall be focused on an illuminated target, placed at a distance of 10.7 meters from the telescope objective, comprising a central dot and a concentric circle. The telescope shall be so aligned that the image of the central dot falls on the intersection of the cross-hairs in the focal plane of the ocular, The plate shall be held in front of the objective lens of the telescope, and if the intersection point of the cross hairs falls without the image of the circle, the prismatic power exceeds 1/16 prism diopter.

4.2.4.8 Refractive power and astigmatism. Test the plates for refractive power and astigmatism in accordance with the applicable test paragraphs of ANSI Z87.1.

4.2.4.9) Flame propagation rate (plastic plates only). Ignite the specimens by holding one end of the specimen horizontally at the top of a luminous 19-mm Bunsen burner flame in a draft-free room. The rate of propagation determined by a stop watch shall be 24 seconds per inch or less.

4.2.4.10 Thermal shock. Test type III plates by hearing them in water at 66° ± 1°C for 3 minutes and then plunging them into water at a temperature of 4“ + loco Laminated glass style a plates shall be tested in water to 86° ± l°C for–3 minutes before plunging into water at a temperature of 4° + 1°C. Check the optical surfaces for delamination, deterioration of optical surfaces, and fracture of glass.

4.2.4.11 Radiant energy transmittance.

4.2.4.11.1 Visible transmittance (styles a, b, and c plates). The standard source of radiant energy used in the measurement of the transmittance of visible radiation (“light”) of eye-protective plates shall be a projection-type lamp No. T-8 *(or* other high-powered, gas-filled, tungsten-filament, electric, incandescent lamp), operated at rated voltage. The visible radiation shall be

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determined photometrically by an observer having normal color vision, as determined by the Holmgren test for color vision, with a physical photometer consisting of a thermopile (or other radiometer) and a luminosity solution having a special transmittance curve which coincides closely with the visibility curve of the average eye, or with a spectrometer applying the standard correction for eye color sensitivity.

4.2.4.11.2 Infrared transmittance (style a plates). The same standard source of radiant energy used in determining the transmittance of visible radiation shall be used also in the measurement of the transmittance of the total infrared radiation. This may be accomplished by observing the infrared spectral energy distribution curves of the gas-filled lamp with and without the style a plates interposed before the entrance slit of the spectrometer, and integrating the area under each of the two curves between the spectral limits 700 and 2,800 nanometers. Any method recognized as efficient by the National Bureau of Standards may be used for determining the infrared transmittance.

4.2.4.11.3 Ultraviolet transmittance (style a plates). Any method recognized as efficient by the National Bureau of Standards may be used for determining the ultraviolet transmittance. If the spectroradiometer is used, its optical parts shall be of quartz or other material transparent to the extreme ultraviolet. To eliminate the effect of stray radiation, especially in the extreme ultraviolet, the shutter which is used in admitting the radiation of the source into the spectroradiometer while taking the zero reading shall be of red or yellow glass that is opaque to ultraviolet radiation. The source of radiant energy for determining the ultraviolet spectral transmittance may be a quartz mercury *arc. The* intense emission lines of the quartz mercury arc at 313, 334, 365 and 405 nanometers are conveniently distributed and well adapted for making these measurements. If other sources are used, the wave lengths closest to the above values of the mercury arc may be used. If a high-powered (2,000- to 3,000-watt), gas-filled, tungsten lamp (which is especially strong in infrared radiation) is used as a source of ultraviolet radiation, special precautions shall be taken to eliminate the effect of stray visible and infrared radiation from the measurements in ultraviolet.

4.2.4.12 Impact resistance. Support style a and style c plates on a rigid frame having internal dimensions of 45 mm by 102 mm. Place a washer of neoprene rubber packing of 40 ± 5 durometer reading, not more than 3.175-mm thick and of the same internal dimensions as the support, between the plate and the support. For style a plates, freely drop a 15.9-mm steel ball, weighing approximately 16 g, from a height of 1 m onto the center of the horizontal outer surface of the plate. Check the plate for fracture from the impact of the steel ball. For style c plates, freely drop a 25.4-mm steel ball, weighing approximately 68 g, from a height of 1.27 meters onto the center of the horizontal outer surface of the plate. Check the plate for fracture from the impact of the steel ball.

4.2.4.13 Maximum transparency. View the plates in transmitted light to determine their dominant hue.

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4.2.4.14 Haze. Test the plastic plates in accordance with ASTM D 1003 to determine haze.

4.2.4.15 Water resistance. Immerse style b plates in water at 20″ ± 2°C for 1 hour. Allow the plates to dry for 24 hours, and clean with lens paper. Any evidence of peeling or other signs of film disintegration shall be cause for rejecting the style b plates.

4,2.4.16 penetration (plastic plates only). Support the plate on a wooden block of such size and shape as to fit the plate securely. A pointed projectile consisting of a new Singer #25 or equal, size 135 by 17 needle, fastened into a holder weighing approximately 44.2 grams, shall be freely dropped, pointed downward, from a height of 1.27 meters onto the horizontal outer surfaces of the plate. The projectile may be guided, but not restricted, in its fall by being dropped through a tube extending to within approximately 102 mm of the plate. Check the plate to determine if it has been pierced through from the impact.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Helmets or handshields shall be cleaned in accordance with method C-1 of MIL-P-116.

5.1.l.2 Drying. Helmets or handshields shall be dried in accordance with MIL-P-116, method optional.

5.1.1.3 Unit packing.

5.1.1.3.1 Helmets or handshields. Helmets *or* handshields shall be individually unit packed in a close-fitting fiberboard box conforming to class weather-resistant, style optional of PPP-B-636.

5.1.1.3.2 Plates. Plates not installed in helmets or handshields shall be individually packaged in kraft paper envelopes and further packaged 12 to a container in a container conforming to PPP-B-566, PPP-B-676, or PPP-B-665. Box closures shall conform to the applicable box specification and the appendix thereto. Plates shall be adequately cushioned to prevent damage during ship ment and handling.

5.1.1.4 Intermediate packing. When the quantity warrents, packages of 12 plates shall be packaged in a fiberboard box conforming to PPP-B-636. Gross weight of intermediate boxes shall not exceed 9.1 kilograms.

5.1.2 Commercial. Helmets, handshields or plates shall be preserved in accordance with ASTM D 3951.

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5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.l Level A. Helmets, handshields *or* plates, unit packed or intermediately packed in accordance with 5.1, shall be packed in overseas type shipping containers conforming to box, wood, cleated, veneer, paper overlaid of pPP-B-576; boxes, wood, wirebound of PPP-B-585; boxes, shipping, fiberboard, wood-cleated of PPP-B-591; boxes, wood, cleated-plywood of PPP-B-601; *or* boxes, wood , nailed and lock-corner of PPP-B-636, at the option of the contractor. Shipping containers shall have caseliners conforming to MIL-L-10547 and shall be closed and sealed in accordance with the appendix thereto. Caseliners for boxes conforming to PPP-P-636 may be omitted provided all joints and corners of the boxes are sealed with tape conforming to PPP-T-76. Box closures shall be as specified in the applicable box specification *or* appendix thereto. The gross weight *of* wood boxes shall not exceed 91 kilograms. Shipping containers shall be closed and reinforced as specified in the applicable box specification.

5.2.2 Level B. Helmets, handshields or plates, intermediately packed in accordance with 5.1 in quantities as specified (see 6.2), shall be packed in domestic type shipping containers conforming to box, wood, cleated, veneer paper overlaid of PPP-B-576; boxes, wood, wirebound of PPP-B-591; boxes, wood, cleated-plywood of PPP-B-621; or boxes, shipping, fiberboard of PPP-B-636, at the option of the contractor.

5.2.3 Commercial. Helmets, handshields or plates, unit packed as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Rough handling. The completed package, packed as for shipment, shall pass the rough handling test specified in MIL-P-116. Unless otherwise specified (see 6.2), the test need not be repeated if the contractor has previously submitted an identical pack acceptable to the inspector.

5.4 Marking.

5.4.1 Civil agencies. Shipments shall be marked in accordance with Fed. Std. No. 123.

5.4.2 Military agencies. Level A and B interior and exterior shipping containers shall be marked in accordance with MIL-STD-129. Commercial packs and unit packs shall be marked in accordance with ASTM D 3951.

5.5 Use of polystyrene (loose-fill) material. Use of polystyrene (loose-fill) material is prohibited.

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6. NOTES

6.1 Intended use.

6.1.1 Helmet and handshield. The general function of the helmet is to protect the face, head, and neck from the hazards present in any gas or electric welding and cutting, while that of the handshield is to protect the face and head.

6.1.2 Style a plates.

6.1.2.1 Selection of shade numbers for welding plates. The following is a guide for the selection of the proper shade numbers of style a plates used in welding:

Welding operation	Suggested shade number
Shielded metal-arc welding	
1/16-, 3/32-, 1/8-, 5/32-inch electrodes	10 or 11
Shielded metal-arc welding	
3/16-, 7/32-, l/4-inch electrodes	12
5/16-, 3/8-inch electrodes	14
Gas tungsten-arc welding	9 to 12
Gas metal-arc welding	9 to 14
Atomic hydrogen welding	10 to 14
Carbon-arc welding	14
Soldering	2
Torch brazing	3 or 4
Light cutting, up to 1 inch	3 or 4
Medium cutting, 1 inch to 6 inches	4 or 5
Heavy cutting, over 6 inches	5 or 6
Gas welding (light), up to 1/8 inch	4 or 5
Gas welding (medium), 1/8 inch to 1/2 inch	5 or 6
Gas welding (heavy), over 1/2 inch	6 or 8

6.1.3 Style b plates. Style b plates are intended as a protection against pitting of the style a plates.

6.1.4 Style c plates. Style c plates are intended to shield the wearers eyes against flying objects.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) Type, style, and class required (see 1.2)
(c) Shade of style a plates required (see 3.11.2.2)
(d) Whether style b plates shall be uncoated or plastic-coated glass or allyl plastic (see 3.11.3.1)
(e) If plates are desired for class 2, lift front (see 3.9.3)

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(f) Level of preservation and packing required (see 5.1 and 5.2)
(g) Quantity per intermediate pack required (see 5.2.2)
(h) Whether rough handling test is required (see 5.3)

MILITARY INTERESTS:

Custodians:

Army - EA
Navy - SH
Air Force - 99

Review activity:

DLA - GS
Air Force - 84

CIVIL AGENCY COORDINATING ACTIVITY:

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Preparing activity:

Army - EA

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
GGG-H-211D	HELMETS, WELDERS'; HANDSHIELD, WELDING; AND PLATES, WELDING

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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